

Endpoint trace-excess direction, shell coanalysis legal reverse, and the shifted-Douglas gap boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-30 · this version issued 2026-07-30 · v1.0

1. Purpose and scope

R-128 left the production root/shell intertwiner, corrected covariance-normal operator bound, and augmented balanced/low gap open. A fresh multi-route audit produces two structural advances and one important route correction.

First, the positive conditional future variance has a precise but one-sided role. It makes the covariance-normal endpoint larger than the R-123 direct trace packet. It therefore does *not* lower-bound that smaller packet and does not bypass the R-123 trace-excess burden. This corrects the initially tempting successor inference before it enters the claim card.

Second, a source-space shell projection is unnecessary for the legal reverse. An orthogonal physical shell analysis, followed by the legal predictable source coanalysis, factors the common source Hessian as

$$H = R^* F = F^* R. \quad (1.1)$$

Once the owner-complete physical response F and its forward block estimate are proved, the aggregate reverse is its true adjoint with exactly the same operator norm. This construction neither assumes nor needs $[\Delta C_b, \Pi_m] = 0$.

Third, the R-127 augmented Douglas criterion admits an exact shifted version which tests a prescribed positive gap μ . It yields a simple norm-level three-channel sufficient condition and makes the remaining low-channel budget quantitative.

The scope is fixed finite cutoff, fixed positive rational floor, one fixed R-093/R-104 temporally faithful chart, its bounded predictable source map, exhaustive orthogonal spatial Fourier shells including the low shell, and a bounded owner-complete physical response on the cylindrical core. The conditional law is fixed when the direct covariance-normal derivative is taken. No production-uniform forward decay, complete balanced estimate, low anchor, common-terminal/matching-trace theorem, $\text{OVERLAP}_{\text{src}}$, Nelson estimate, removal, interacting measure, or Sector-A closure is claimed.

2. Exact endpoint direction and the failed action transfer

For one exhaustive owner-complete conditional packet put

$$\Phi = \mathbb{E}_f J, \quad \mathcal{V} = \mathbb{E}_f \|J - \Phi\|^2, \quad \mathcal{T} = \Theta - \|\Phi\|^2. \quad (2.1)$$

The R-123 direct trace packet and the R-125 covariance-normal endpoint are

$$\mathcal{P}_{\text{comp}} = -\frac{1}{2}\mathcal{T}, \quad \mathcal{E}_{\text{CN}} = \frac{1}{2}(\mathcal{V} - \mathcal{T}). \quad (2.2)$$

Conditional Pythagoras gives

$$\boxed{\mathcal{E}_{\text{CN}} = \frac{1}{2}(\mathbb{E}_f \|J\|^2 - \Theta) = \mathcal{P}_{\text{comp}} + \frac{1}{2}\mathcal{V} \geq \mathcal{P}_{\text{comp}}.} \quad (2.3)$$

Equation (2.3) is an absolute conditional-endpoint identity. Its direction does not give a lower bound on $\mathcal{P}_{\text{comp}}$ from a lower bound on $\mathcal{E}_{\text{CN}}$. The R-125 constant-translation owner makes the issue exact: for $s > 0$,

$$\mathcal{V} = 4s, \quad \mathcal{T} = 4s, \quad \boxed{\mathcal{E}_{\text{CN}} = 0, \quad \mathcal{P}_{\text{comp}} = -2s.} \quad (2.4)$$

Thus the larger endpoint can be harmless while the actual R-123 packet is strictly negative.

The endpoint comparison is also not stable under subtraction. With $J_0 = \eta$, $J_h = 0$, $\mathbb{E}\eta = 0$, $\mathbb{E}\eta^2 = 1$, and $\Theta_0 = \Theta_h = 0$,

$$\Delta\mathcal{E}_{\text{CN}} = -\frac{1}{2}, \quad \Delta\mathcal{P}_{\text{comp}} = 0. \quad (2.5)$$

Hence neither a moving-endpoint secant nor a stationary subtraction may drop $\Delta\mathcal{V}$ by positivity.

Theorem 2.1 (endpoint-direction firewall)

At an absolute conditional endpoint, (2.3) holds. It supplies no lower bound for the R-123 action owner $\mathcal{P}_{\text{comp}}$ and no signed statement for an endpoint difference. The production Nelson route therefore still requires the aggregate R-123 estimate

$$\boxed{\sum_{k,C} \mathbb{E}[\Theta_{k,C} - \|\Phi_{k,C}\|^2] \leq 2\eta X + 2\zeta Y + 2C, \quad \eta < \frac{9}{20}, \quad \zeta < \frac{3}{20}.} \quad (2.6)$$

R-129 verifies only the direction and the acceptance range. It does not prove the production hypothesis (2.6).

3. Direct signed covariance-normal differentiation

Although future-variance positivity does not repair the R-123 action, the algebraic Hessian audit still removes an unnecessary methodological restriction. At a fixed conditional law, assume J and Θ are C^2 and differentiation commutes with $\mathbb{E}_f$. Write

$$\dot{J}_H = DJ[H], \quad \ddot{J}_{H,K} = D^2J[H, K]. \quad (3.1)$$

Then

$$\boxed{D\mathcal{E}_{\text{CN}}[H] = \text{Re } \mathbb{E}_f \langle J, \dot{J}_H \rangle - \frac{1}{2}D\Theta[H],} \quad (3.2)$$

and

$$\boxed{D^2\mathcal{E}_{\text{CN}}[H, K] = \text{Re } \mathbb{E}_f \{ \langle \dot{J}_H, \dot{J}_K \rangle + \langle J, \ddot{J}_{H,K} \rangle \} - \frac{1}{2}D^2\Theta[H, K].} \quad (3.3)$$

All derivatives of $\Phi = \mathbb{E}_f J$ cancel. Formula (3.3) retains the full nonnegative derivative Gram on the diagonal and one signed curvature remainder. It does not by itself bound that remainder.

The loss from separately estimating $D^2\mathcal{V}$ and $D^2\mathcal{T}$ can be arbitrarily large. Let $\eta_0, \eta_1, \eta_0^\circ$ be independent standard Gaussians and put

$$J_{n,z} = (\sqrt{n} z \eta_0, z \eta_1), \quad J_{n,z}^\circ = (\sqrt{n} z \eta_0^\circ, 0), \quad \Theta_{n,z} = n z^2. \quad (3.4)$$

Then

$$\mathcal{V}_{n,z} = (n+1)z^2, \quad \mathcal{T}_{n,z} = n z^2, \quad \mathcal{E}_{\text{CN},n}(z) = \frac{z^2}{2}, \quad (3.5)$$

so

$$D^2\mathcal{V}_n = 2n + 2, \quad D^2\mathcal{T}_n = 2n, \quad \boxed{D^2\mathcal{E}_{\text{CN},n} = 1.} \quad (3.6)$$

Thus separate uniform norm bounds are not logically necessary *if* a direct owner-complete signed estimate is proved. The fixture is methodological and is not an A1 production counterexample.

4. Analytic shortcuts tested and rejected

4.1 Conditional Poincare does not give parameter semiconvexity

For a standard Gaussian η and

$$J_N(z, \eta) = \cos(Nz)\eta, \quad (4.1)$$

the Gaussian Poincare inequality is an equality:

$$\text{Var}(J_N) = \cos^2(Nz) = \mathbb{E}|D_\eta J_N|^2. \quad (4.2)$$

Nevertheless

$$\frac{d^2}{dz^2} \text{Var}(J_N)(0) = -2N^2. \quad (4.3)$$

Differentiating a Poincare upper bound therefore cannot supply a uniform parameter Hessian bound. This is an abstract diagnostic, not a production counterexample.

4.2 Gaussian second score is exact but moment-expensive

For deterministic or strict-past u, v and sufficiently integrable F , second Gaussian integration by parts gives

$$\mathbb{E}D_\xi^2 F[u, v] = \mathbb{E}[(\langle \xi, u \rangle \langle \xi, v \rangle - \langle u, v \rangle) F]. \quad (4.4)$$

The score has exact L^2 norm

$$\sqrt{\|u\|^2 \|v\|^2 + \langle u, v \rangle^2} \leq \sqrt{2} \|u\| \|v\|, \quad (4.5)$$

and $\sqrt{2}$ is sharp for parallel vectors. The route removes explicit feedback jets only after complete Gaussian integration, but it demands an L^2 endpoint bound. On the R-122 coherent-amplitude family the relevant current-square endpoint reaches the eighth moment $\mathbb{E}A_t^8 = e^{8t^2}$. No registered source/sextic graph budget controls that separated moment uniformly. The route is therefore parked unless a complete signed endpoint supplies the missing cancellation before Cauchy–Schwarz.

4.3 Entropy does not control the second score

Let γ be the standard Gaussian law on $\mathbb{R}$ and

$$\rho_N(x) = 1 + \epsilon \sin(Nx), \quad 0 < \epsilon < 1. \quad (4.6)$$

Then $\int \rho_N d\gamma = 1$ and the elementary inequality $(1 + u) \log(1 + u) \leq u + u^2$ gives

$$\text{Ent}_\gamma(\rho_N) \leq \frac{\epsilon^2}{2}. \quad (4.7)$$

The Fisher information instead obeys

$$I_\gamma(\rho_N) = \epsilon^2 N^2 \mathbb{E}_\gamma \frac{\cos^2(NX)}{1 + \epsilon \sin(NX)} \geq \frac{\epsilon^2 N^2 (1 + e^{-2N^2})}{2(1 + \epsilon)}. \quad (4.8)$$

It diverges quadratically. Normalized entropy and Doob orthogonality cannot control the conditional Fisher or second-score terms; for a Gibbs density those terms contain the force and Hessian being sought and the argument is circular.

4.4 A rational second derivative is not separately floor-uniform

For a homogeneous rational A1 row, scaling the floor $e > 0$ at $u = \sqrt{e}v$ gives

$$c_e(\sqrt{e}v) = \sqrt{e}c_1(v), \quad Dc_e(\sqrt{e}v) = Dc_1(v), \quad D^2c_e(\sqrt{e}v) = e^{-1/2}D^2c_1(v). \quad (4.9)$$

An unweighted D^2c supremum is therefore not floor-uniform. The complete products cD^2c and $(Dc)^2$ are scale-neutral and must remain inside the signed owner-complete Hessian.

5. Total covariance does not intertwine temporal increments

R-120 proves that the total regulated covariance C_J is a Fourier multiplier. That fact does not imply that each temporal increment in an arbitrary faithful chart commutes with a spatial shell. Let

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Delta C_1 = P_+ = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Delta C_2 = P_- = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (5.1)$$

Both $P_{\pm}$ are orthogonal projections, $P_+ + P_- = I$, and the total covariance commutes with P . However

$$P_+P = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \neq \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = PP_+. \quad (5.2)$$

Since $(\Delta C_b)^{1/2} = \Delta C_b$, the block synthesis does not intertwine the componentwise source shell. This fires the temporal-increment commutation no-go below. It does not refute a separately pinned canonical spectral covariance path. More importantly, Section 6 avoids the assumption entirely.

6. Shell analysis/coanalysis legal-reverse theorem

Let

$$\mathcal{S}_{\pi}^{\text{pred}} = \bigoplus_{b=1}^B L^2(\Omega, \mathcal{F}_{t_{b-1}}; \mathcal{S}_b), \quad Lh = \sum_b S_b h_b \quad (6.1)$$

be the R-127 predictable source map into the physical Hilbert space $\mathcal{X}$. Its legal adjoint is

$$(L_{\text{pred}}^* G)_b = \mathbb{E}[S_b^* G \mid \mathcal{F}_{t_{b-1}}]. \quad (6.2)$$

Let $\{\Pi_m\}_{m \geq \ell}$ be an exhaustive mutually orthogonal deterministic spatial Fourier-shell resolution, including the low shell. Put

$$\mathcal{Y} = \bigoplus_m \text{Ran } \Pi_m, \quad \mathcal{C}z = (\Pi_m z)_m. \quad (6.3)$$

Then $\mathcal{C}$ is an isometry and $\mathcal{C}^* \mathcal{C} = I$.

Assume that the fully recombined physical Hessian response has been defined as a bounded map

$$A : \mathcal{S}_{\pi}^{\text{pred}} \rightarrow \mathcal{X}, \quad (6.4)$$

and that its legal source pullback

$$H = L^* A \quad (6.5)$$

is selfadjoint. This is an assumption on the owner-complete response, not a claim that the R-126 displayed force has already been identified with it. Define

$$R = \mathcal{C}L, \quad F = \mathcal{C}A. \quad (6.6)$$

Theorem 6.1 (physical-shell analysis and legal source coanalysis)

Under (6.1)–(6.6),

$$\boxed{R^*F = L^*A = H, \quad F^*R = A^*L = H.} \quad (6.7)$$

Moreover

$$\boxed{\|Rh\|_Y^2 = \sum_m \|\Pi_m Lh\|^2 = \|Lh\|^2, \quad \|R\| = \|L\|,} \quad (6.8)$$

and for $y = (y_m)_m \in \mathcal{Y}$,

$$\boxed{\left\| \sum_m L^* \Pi_m y_m \right\| \leq \|L\| \left(\sum_m \|y_m\|^2 \right)^{1/2}.} \quad (6.9)$$

Every receiving source block in (6.9) contains the conditional expectation in (6.2), so the coanalysis is predictable and legal.

Proof. The isometry gives $R^*F = L^*C^*CA = L^*A = H$. Taking adjoints and using $H = H^*$ gives $F^*R = H$. Parseval proves (6.8), and the adjoint operator norm proves (6.9). No commutator between S_b and Π_m occurs. $\square$

Let E_b be the temporal source projections. The rectangular forward cells and the corresponding pulled cells are

$$F_{mb} = \Pi_m A E_b, \quad T_{mb} = L^* \Pi_m A E_b. \quad (6.10)$$

Their true adjoints are

$$F_{mb}^* = E_b A^* \Pi_m, \quad T_{mb}^* = E_b A^* \Pi_m L. \quad (6.11)$$

The second orientation is predictable because it ends in a source block; it need not equal an unproved swapped-label geometric cell. Strong summation gives

$$\sum_{m,b} T_{mb} = H, \quad \sum_{m,b} T_{mb}^* = H. \quad (6.12)$$

The quadratic owner is written once as

$$\text{Re}\langle Rh, Fh \rangle = \frac{1}{2} \langle h, (R^*F + F^*R)h \rangle = \langle h, Hh \rangle. \quad (6.13)$$

Writing both full sums without the factor 1/2 would double the owner.

The distinction between a true adjoint orientation and a swapped geometric label is exact. Take

$$L = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \Pi_i = E_i, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}. \quad (6.14)$$

For $T_{mb} = L^* \Pi_m B L E_b$, direct multiplication gives $T_{21} = 0$ while $T_{12} \neq 0$. Hence

$$T_{21}^* = 0 \neq T_{12}. \quad (6.15)$$

The theorem controls the adjoint of a proved forward region. It does not reflect that region into a distinct lower geometric band by exchanging indices. Likewise $L^* \Pi_1 e_1 = 2^{-1/2}(1, 1)$ occupies both source coordinates, so source-shell localization must not be asserted after coanalysis. Weighted Schur is executed in the orthogonal physical shell space before applying R^* .

Corollary 6.2 (forward estimate transfers to the aggregate reverse)

If an owner-complete weighted Schur argument proves

$$\|F\| \leq K_F, \quad (6.16)$$

then

$$\boxed{\|F^*\| = \|F\| \leq K_F, \quad \|H\| \leq \|L\|K_F.} \quad (6.17)$$

The currently available uniform upper bound is $\|L_{\text{pred}}\|^2 \leq \|C_J\|$; equality is not automatic because the blockwise conditional expectations can contract the source adjoint.

Equation (6.17) does not import an old physical forward constant unchanged after pullback. The scalar fixture

$$Lx = (2x, 0), \quad \Pi = \text{diag}(1, 0), \quad A = L \quad (6.18)$$

has physical forward norm 2 and pulled norm 4. The synthesis factor is real and must be budgeted or removed by a normalized quotient theorem.

Nor does the source Hessian alone canonically determine a shell split. Let

$$Lx = \frac{x}{\sqrt{2}}(1, 1), \quad B_1 = I, \quad B_2 = \text{diag}(2, 0). \quad (6.19)$$

Then $L^*B_1L = L^*B_2L = 1$, but for $\Pi = \text{diag}(1, 0)$,

$$L^*\Pi B_1L = \frac{1}{2}, \quad L^*\Pi B_2L = 1. \quad (6.20)$$

The new theorem therefore requires the owner-complete physical response A , not merely an abstract pulled Hessian H .

For smooth Littlewood–Paley cutoffs, (6.3) must be replaced by explicit analysis/coanalysis frame bounds and a dual reconstruction. A partition of unity alone is not Parseval: the scalar choice $\Pi_1 = \Pi_2 = I/2$ has $\sum_m \Pi_m = I$ but $\sum_m \Pi_m^* \Pi_m = I/2$. The theorem uses the already admitted orthogonal Fourier-shell convention.

There is also a quotient/refinement boundary. For $L(h_1, h_2) = h_1 + h_2$, the vector $(1, -1)$ lies in $\text{Ker } L$, but neither L^*LE_1 nor L^*LE_2 kills it. Only their sum does. Individual temporal cells are refinement-basic only under an extra condition such as $E_b \text{Ker } L \subseteq \text{Ker } L$; otherwise children must be aggregated before a quotient or representation-preserving refinement assertion is made.

7. Exact shifted-Douglas uniform-gap theorem

Use the R-128 full-cross convention and put

$$E = 2\eta I - R_0, \quad F_0 = 2\zeta I - S_0, \quad D = -L_0, \quad (7.1)$$

$$M_2 = \begin{pmatrix} E & -A_{\text{eff}}/2 \\ -A_{\text{eff}}^*/2 & F_0 \end{pmatrix}, \quad K = \begin{pmatrix} B \\ C \end{pmatrix}, \quad \mathcal{M} = \begin{pmatrix} M_2 & -K \\ -K^* & D \end{pmatrix}. \quad (7.2)$$

Theorem 7.1 (shifted Douglas criterion)

For $\mu \geq 0$, define

$$D_\mu = D - \mu I. \quad (7.3)$$

Then

$$\boxed{\mathcal{M} \succeq \mu I} \quad (7.4)$$

if and only if

$$\boxed{D_\mu \succeq 0, \quad \text{Ran } K^* \subseteq \text{Ran } D_\mu^{1/2}, \quad M_2 - \mu I - X_\mu^* X_\mu \succeq 0,} \quad (7.5)$$

where X_μ is the Douglas reduced solution of

$$D_\mu^{1/2} X_\mu = K^*. \quad (7.6)$$

A Moore–Penrose expression may be used only after closed range is proved.

Proof. Subtract μI from the three-channel matrix and apply the bounded-operator Douglas–Schur theorem to the lower-right block D_μ . $\square$

Corollary 7.2 (norm-level three-channel gap)

Assume

$$E \succeq eI, \quad F_0 \succeq fI, \quad \|A_{\text{eff}}\| \leq a, \quad D \succeq dI, \quad \|K\| \leq k. \quad (7.7)$$

The two-channel lower bound is

$$m_2 = \frac{e + f - \sqrt{(e - f)^2 + a^2}}{2}. \quad (7.8)$$

If

$$a^2 < 4ef, \quad k^2 < dm_2, \quad (7.9)$$

then

$$\boxed{\mathcal{M} \succeq \mu_3 I, \quad \mu_3 = \frac{m_2 + d - \sqrt{(m_2 - d)^2 + 4k^2}}{2} > 0.} \quad (7.10)$$

This follows by first replacing M_2 by $m_2 I$ and then taking the least eigenvalue of the remaining scalar two-by-two matrix.

For a desired $0 \leq \mu < d$, put

$$\tau_\mu = \mu + \frac{k^2}{d - \mu}. \quad (7.11)$$

The explicit sufficient conditions

$$e > \tau_\mu, \quad f > \tau_\mu, \quad a < 2\sqrt{(e - \tau_\mu)(f - \tau_\mu)} \quad (7.12)$$

imply (7.4).

At the R-128 zero-diagonal half-debt specialization,

$$m_2 = 0.02396011633137279 \dots \quad (7.13)$$

Consequently a low reserve $D \succeq dI$ and coupling $\|K\| \leq k$ satisfy the simple sufficient threshold

$$\boxed{k^2 < dm_2.} \quad (7.14)$$

The number (7.13) is not a general production margin: it assumes $R_0 = S_0 = 0$ or that those adverse diagonal debts were already absorbed.

In the scalar three-channel case the exact Schur determinant reads

$$\boxed{\left(2\eta - \frac{b^2}{d}\right) \left(2\zeta - \frac{c^2}{d}\right) > \left(\frac{a}{2} + \frac{bc}{d}\right)^2.} \quad (7.15)$$

Together with $d > 0$ and the two Schur diagonal conditions

$$2\eta - \frac{b^2}{d} \geq 0, \quad 2\zeta - \frac{c^2}{d} \geq 0, \quad (7.16)$$

the non-strict version of (7.15) is the scalar PSD criterion; strict diagonals and strict (7.15) give positive definiteness. The determinant exhibits the correlation between the two low couplings; separate triangle payments can be strictly worse.

8. Strongest current production criterion

The exact live burden remains (2.6). The available pieces may be organized, conditionally, as follows. Let

$$X = \mathbb{E} \sum_k \|h_k\|^2, \quad Y = \mathbb{E} \|Z_K\|_6^6, \quad \kappa^2 = \frac{6}{125P}. \quad (8.1)$$

R-123's one genuine adapted row gives

$$\mathcal{T}_{\text{row}} \leq 2\eta_r X + 2\zeta_r Y + \frac{\kappa^4}{8\sqrt{\eta_r \zeta_r}}. \quad (8.2)$$

If the still-unproved owner-complete physical response has forward constants $C_{\text{mix}}, C_{\text{far}}$ in the R-127 hypotheses, the sharp registered aggregation factor is

$$K_{\text{tri}} = \sqrt{\frac{8}{7}} C_{\text{mix}} 2^{-j_0-2C} + \sqrt{\frac{224}{223}} C_{\text{far}} 2^{-3j_0-4C}. \quad (8.3)$$

For any $\eta_t > 0$ its two-channel Young payment is

$$\mathcal{T}_{\text{tri}} \leq 2\eta_t X + \frac{K_{\text{tri}}^2}{8\eta_t} Y. \quad (8.4)$$

Equivalently $K_{\text{tri}} \leq 4\sqrt{\eta_t \zeta_t}$. These are acceptance tests after the production block theorem, not values of the missing constants.

The balanced collar is scaling-critical. If its action-normalized complete coefficient is c_{bal} , the circle fixture forces

$$\eta_b \zeta_b \geq \frac{c_{\text{bal}}^2}{4}. \quad (8.5)$$

The production c_{bal} is not registered. The low block must satisfy Theorem 7.1, including its range condition, and the absolute low/injected anchor remains required.

Thus even after the legal-reverse repair the missing data are concrete:

$$\boxed{C_{\text{mix}}, C_{\text{far}}, c_{\text{bal}}, R_0, S_0, d, k, \text{ and the absolute low/common-terminal owner.}} \quad (8.6)$$

Every value must be obtained after all six rows, visits, output clusters, heat, rational multipliers, R-063 forest terms, and the 2680/729 Cartan block have been assembled once. No per-row, per-root, per-shell, or refinement-depth constant may be introduced.

9. Evidence-map roadmap

The route decisions in this checkpoint are:

Route	Verdict	Reusable boundary
$\mathcal{E}_{\text{CN}} \geq \mathcal{P}_{\text{comp}}$	advanced identity	absolute endpoint only
Transfer that bound to R-123 action	failed	$\mathcal{P}_{\text{comp}}$ is the smaller owner
Direct signed CN Hessian	advanced algebra	mean derivatives cancel; production bound open
Separate $H_{\mathcal{V}}, H_{\mathcal{T}}$ necessity	failed	the n -fixture has bounded signed difference
Differentiate conditional Poincare	failed	parameter Hessian grows as N^2
Gaussian second score	parked	$\sqrt{2}$ is sharp; endpoint L^2 is unavailable
Entropy/Doob second-score control	failed	bounded entropy, divergent Fisher information
Total-covariance shell inter-twiner	failed	C_J says nothing blockwise
Physical shell analysis/source coanalysis	advanced conditionally	legal reverse follows from the forward response
Shifted Douglas gap	advanced	exact μ -criterion and norm corollary.

Substantive route decisions, including the unsuccessful first interpretation of future-variance positivity, are preserved in the append-only exploration ledger. The exact broad false implications are promoted to the negative registry. Token-by-token private search is not recorded; the reusable question, finite check, finding, boundary, evidence, and revisit condition are.

10. Devil’s-advocate review

1. **Objection: positive future variance can be discarded from the R-123 action. UPHELD against the claim.** Equation (2.4) fixes the direction: the action owner is $\mathcal{P}_{\text{comp}} = \mathcal{E}_{\text{CN}} - \mathcal{V}/2$, so the variance is adverse when recovering that owner.
2. **Objection: the absolute endpoint identity also holds for a secant. UPHELD against the claim.** Equation (2.5) gives an exact counterexample. R-125’s complement leakage must also be retained when the output resolution is not exhaustive.
3. **Objection: the direct signed Hessian closes production because the conditional mean disappears. UPHELD against the overclaim.** The derivative Gram and the signed $J\ddot{J} - D^2\Theta/2$ remainder remain.
4. **Objection: conditional Poincare bounds the variance Hessian. UPHELD against the claim.** Equations (4.1)–(4.3) separate the Gaussian derivative coordinate from the external parameter derivative.
5. **Objection: total Fourier-diagonal covariance makes every temporal source block Fourier diagonal. UPHELD against the claim.** The $P_{\pm}$ decomposition has total identity covariance and noncommuting increments.
6. **Objection: Theorem 6.1 silently assumes the temporal commutator. DISMISSED.** It applies Π_m only after L and uses $L^*\Pi_m$ as the true legal coanalysis. No projector is moved through S_b .
7. **Objection: the coanalysis makes the old forward constants production constants. UPHELD against the overclaim.** The forward response A , its complete ownership, and its physical block constants remain unproved; pullback also costs $\|L\|$ unless normalized.

8. **Objection: forward and reverse are two favorable owners. UPHELD against the claim.** Equation (6.13) counts the symmetrized pair once. Adding both complete sums without $1/2$ doubles H .
9. **Objection: $m_2 = 0.0239601 \dots$ pays any low block. UPHELD against the claim.** It is a zero-diagonal or already-paid specialization. The production D, K, R_0, S_0 and Douglas range condition are still missing.
10. **Objection: R-129 closes an A13 gate. DISMISSED.** It closes only a structural legal-reverse mechanism conditional on the physical response and supplies acceptance criteria. The forward, balanced, low, anchor, $\text{OVERLAP}_{\text{src}}$, and Nelson burdens remain.

11. Reproduction commands

Primary symbolic and exact audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/
a13_classii_endpoint_trace_excess_shell_coanalysis_
shifted_douglas_boundary.py
```

Non-importing standard-library audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/
a13_classii_endpoint_trace_excess_shell_coanalysis_
shifted_douglas_boundary_independent.py
```

Integrated authority, PDF, exploration, registry, and public-surface audit:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/
a13_classii_endpoint_trace_excess_shell_coanalysis_
shifted_douglas_boundary_verify.py
```

12. Result footer

Result ID: A13-CLASSII-ENDPOINT-TRACE-EXCESS-SHELL-COANALYSIS-SHIFTED-DOUGLAS-BOUNDARY (R-129).

Precise statement: endpoint direction and transfer firewall in Theorem 2.1; direct signed Hessian in Eqs. (3.2)-(3.6); analytic shortcut boundaries in Section 4; temporal increment no-go in Section 5; shell analysis/coanalysis Theorem 6.1 and Corollary 6.2; shifted-Douglas Theorem 7.1 and Corollary 7.2; production acceptance map in Section 8.

Scope: fixed cutoff and floor; fixed temporally faithful chart and conditional law; bounded predictable source synthesis; exhaustive orthogonal spatial shells; bounded owner-complete physical response on the cylindrical core.

Dependencies: A1; R-093, R-103, R-104, R-119, R-120, R-122--R-128.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_endpoint_trace_excess_shell_coanalysis_
shifted_douglas_boundary_verify.py

Expected output: R-129 integrated PASS; primary and independent assertion counts, manifest, PDF, exploration, negative, and public-surface gates pass.

Falsification gate: failure of endpoint Pythagoras or owner direction; direct signed derivative cancellation; separate-norm, Poincare, entropy/Fisher, temporal-increment, response-factorization, pullback-factor, or ambient-split fixture; shifted Douglas identity, low threshold, authority/hash,

PDF, exploration, registry, or public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no transfer of E_CN dominance to the R-123 action; no production physical response, forward block constants, balanced/low closure, absolute anchor, common terminal, OVERLAP_src, Nelson, removal, interacting measure, Sector-A closure, or T5--T7 follows.
Next required action: prove A13-CLASSII-OWNER-COMPLETE-TRACE-EXCESS-PHYSICAL-RESPONSE-FORWARD-BALANCED-LOW-BOUND, including C_mix, C_far, c_bal, the full diagonal/low Douglas data, and the absolute owner once.